

Edwin Alex Vinu

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PROFILE

First-Class Computer Science with Artificial Intelligence graduate from Coventry University with a strong interest in practical engineering and automation. Currently working as an Technical Quality Assurance (AI Logic), focused on **technical verification and logic testing**. Proven ability to handle complex projects, such as designing an AI-based sorting system and leading a technical team. Seeking a graduate role in **Systems, Controls, or Technical Operations** where I can apply my logical foundation to physical hardware and help solve complex engineering challenges.

WORK EXPERIENCE

Outlier AI

Remote

Technical Quality Assurance (AI Logic)

Oct 2025 - Present

- Technical Quality Assurance:** Perform rigorous verification of complex system outputs, ensuring 100% alignment with strict logic, accuracy, and safety benchmarks.
- Systematic Troubleshooting:** Identify and document technical bugs and edge-case failures using a structured analytical approach to improve system reliability.
- Data Integrity Management:** Execute high-volume technical tasks under tight deadlines, maintaining accountability and high standards of detail.
- Logic Testing:** Develop and refine test cases to stress-test system responses, applying the same analytical rigour used in industrial systems evaluation.

PROJECT EXPERIENCE

Coventry University

Scrum Project - 1st Class Grade

Jan 2025 - May 2025

- Led a six-person Python development team using **Agile/Scrum methodology**, **cutting app load time by 60%** and improving user interaction by **65%**
- Managed sprint planning, code reviews, and delivery within a three-month deadline — awarded a 1st class grade

AI-Based E-Waste Sorting System

Sep 2024 - May 2025

- Designed and built a **Python computer vision system** using **custom CNN and MobileNetV2 deep learning models** to autonomously identify and classify electronic waste in real time
- Implemented rigorous **data preprocessing pipelines, validation, and error handling** to ensure robust model performance with real-world inputs
- Benchmarked model accuracy using **quantitative metrics**, applied GitHub **version control**, and delivered a fully documented end-to-end solution meeting **ethical and sustainability requirements**

Hospital Readmission Prediction - Machine Learning

Jan 2025 - May 2025

- Built and evaluated **three ML models in Python across a 70,000+ patient dataset**, improving prediction precision from **75% to 90%** through **systematic feature engineering** and hyperparameter tuning
- Compared algorithm performance across multiple approaches using quantitative evaluation metrics, identifying the most accurate and reliable solution
- Identified key clinical risk factors influencing hospital readmission rates, producing data-driven insights with potential real-world healthcare decision-making impact

EDUCATION

Coventry University

First Class Honours

BSc Computer Science with Artificial Intelligence

Sep 2022 - Jul 2025

- Achieved **First Class Honours** across a challenging three-year programme covering programming, software engineering, systems design, AI, mathematics, and data analysis
- Developed and delivered complex technical projects from scratch, demonstrating end-to-end planning and problem-solving.

Kettering Science Academy

A-Levels: Mathematics, Business, Computer Science

Sep 2020 - Aug 2022

- EPQ:** 'What are Quantum Computers' - Independent research project demonstrating technical writing and self-directed study

SKILLS & INTERESTS

Languages/Skills: Python, C++, SQL, VS Code, Docker, MySQL, GitHub, System Testing, **Quality Assurance**, Neural Networks, Prompt Engineering

Certificates: ICDL Level 2 Microsoft 365, ICDL Level 3 Spreadsheet, CyberFirst Security